



Polymerizable C₇₀ derivatives with acrylate functionality for efficient and stable solar cells

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ABSTRACT

Fullerene-based organic solar cells are generally suffering from severe microstructure evolution occurring in their bulk heterojunction active layers and thus are extremely stable. To address it, four polymerizable C₇₀ fullerene derivatives, [6,6]-phenyl-C₇₁-ethyl acrylate (PC₇₁EA), [6,6]-phenyl-C₇₁-propyl acrylate (PC₇₁PrA), [6,6]-phenyl-C₇₁-butyl acrylate (PC₇₁BA), and [6,6]-phenyl-C₇₁-pentyl acrylate (PC₇₁PeA), have been designed, synthesized, and investigated. These fullerene compounds have a molecular structure, shape and size very like the conventional C₇₀ fullerene acceptor, [6,6]-phenyl-C₇₁-butyric acid methyl ester (PC₇₁BM), and have been found no different in their light absorption, redox potentials, and frontier orbital energy levels. Using these fullerene acrylates individually as acceptor and poly(3-hexylthiophene) as donor, organic solar cells have been fabricated and gave optimal efficiencies ranging from 3.32% to 4.16%, comparable to PC₇₁BM-based reference cells (4.06%). Owing to their acrylate functionality, these fullerene derivatives can turn into insoluble upon heating, and thus endow their solar cell devices much better thermostability than PC₇₁BM-based reference cells. The best one, coming from PC₇₁PeA devices, reported an optimal efficiency of 4.16%, and maintained 91.7% efficiency after heat treatment at 150 °C for 35 h. As a sharp contrast, the PC₇₁BM reference cell dropped its optimal efficiency from 4.06% to 0.48% only after 5 h heat treatment. X-ray diffraction, optical and atomic force microscopy, and space-charge-limited current method have been carried out to understand active layer structure, morphology, and charge mobility change during heat treatment.

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1. Introduction

Organic photovoltaics (OPVs) have received considerable attention for their potential in producing lightweight, low cost and flexible solar cells [1–3]. In recent years, a big progress has been achieved in device power conversion efficiency (PCE), in which the champion record of single junction devices is over 15% [4,5], while

that of tandem cells exceeds 17% [6]. Although device efficiency needs to be further improved, the OPV practical application requires more effort on addressing device stability issues.

Because of low dielectric constants for most organic materials, the most efficient OPV device structure developed so far comprises of a bulk-heterojunction (BHJ) active layer formed by physically blending at least a photovoltaic donor (D) and a photovoltaic acceptor (A) material [7]. Compared with planar heterojunction, BHJ can provide a large D/A interface and thus improve exciton migration and dissociation efficiencies. However, it also complicates the active layer microstructure and brings a structural evolution problem, an inherent account for device instability. According to light-to-electricity conversion mechanism, the idea microstructure of BHJ active layer is a kind of D/A bicontinuous interpenetrating network with a phase separation size of tens

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